



Representation of Convectively Coupled Equatorial Waves (CCEWs) During Boreal Summer in IITM-IMD GFS T1534 Model Reforecast

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Abstract—Convectively coupled equatorial waves (CCEWs) play a key role in organizing tropical convection, modulating precipitation variability, and influencing sub seasonal predictability. Their realistic representation remains a major challenge for numerical weather prediction models. This study evaluates the representation and lead-time evolution of CCEWs in the Indian Institute of Tropical Meteorology—India Meteorological Department Global Forecast System (IITM-IMD GFS), which currently serves as one of the operational models at IMD since 2018 for short to medium range weather prediction in India. The analysis uses 19 years (2000–2018) of high-resolution (T1534; ~ 12.5 km) reforecast data. Model reforecasts at lead times of Day-1, Day-3, and Day-5 are compared with observations from NOAA outgoing longwave radiation (OLR) and NASA Integrated Multi-satellite Retrievals for GPM (IMERG) precipitation during boreal summer. The model reproduces the broad spatial distribution of tropical convection and precipitation; however, systematic biases are evident in the magnitude and variability of rainfall. In particular, the model tends to overestimate light to moderate rainfall while underestimating heavy rainfall at longer lead times. The filtered high-frequency (2–20-day) OLR variability is substantially underestimated across the tropics, with the suppression becoming more pronounced from Day-1 to Day-5 forecasts. Space–time spectral diagnostics reveal that the model captures the spatial structure of Kelvin, equatorial Rossby (ER) waves and MJO but exhibits biases in their amplitude and variability. Domain-averaged root mean square error (RMSE) of these waves confirms a progressive increase in forecast errors with lead time. This study highlights the deterioration of tropical wave variability with forecast lead time and emphasize the importance of improving the representation of convection-wave interactions for maintaining realistic tropical variability and enhancing monsoon prediction in operational forecast systems.

Keywords: Convectively coupled equatorial waves (CCEWs), Madden–Julian oscillation (MJO), boreal summer, global forecast system (GFS) T1534, reforecast.

1. Introduction

Atmospheric variability in the tropics on daily to intraseasonal timescales is largely governed by the dynamics of Convectively Coupled Equatorial Waves (CCEWs) (Wheeler & Kiladis, 1999). These waves, including Kelvin, Equatorial Rossby (ER), Mixed Rossby-Gravity (MRG), and Tropical Depression (TD)-type waves, are primarily generated by diabatic heating associated with organized tropical convection (Gill, 1980; Kiladis et al., 2009; Wheeler & Kiladis, 1999). Once coupled with convection, these wave modes become dynamically active and are capable of transporting energy, momentum, and moisture across vast longitudinal scales (Yang et al., 2007). CCEWs play a crucial role in modulating tropical convection and rainfall variability, often acting as precursors to larger-scale tropical disturbances (Kiladis et al., 2009; Straub & Kiladis, 2002; Wheeler & Kiladis, 1999). These waves are typically observed between 20°N and 20°S, with maximum amplitudes confined closer to the equator (10°N–10°S), and extend vertically from the lower troposphere to the lower stratosphere. Their propagation characteristics vary depending on wave type: Kelvin waves exhibit fast eastward movement, while ER, MRG, and TD-type waves propagate westward at slower speeds. The propagation of CCEWs can trigger convective storms over large distances, linking weather systems across the tropics and contributing significantly to the spatial and temporal coherence of precipitation (Kiladis et al., 2009).

The existence of CCEWs was first distinctly identified by Wheeler and Kiladis (1999) through satellite-based analyses of Outgoing Longwave

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Radiation (OLR), which revealed their characteristic space–time spectral signatures corresponding to different equatorial wave modes. Matsuno (1966) formulated the shallow-water equations on an equatorial β -plane and demonstrated that the resulting spectrum of equatorially trapped waves—including Kelvin waves, Rossby waves, mixed Rossby-gravity waves, and inertia-gravity waves provides the theoretical foundation for understanding convectively coupled equatorial disturbances. Observational evidence for MRG waves (Maruyama, 1967; Yanai & Maruyama, 1966) and Kelvin waves (Wallace & Kousky, 1968) in the equatorial stratosphere was first identified in the 1960s. Consistent with theoretical expectations, MRG waves are predominantly associated with meridional fluctuations, whereas the Kelvin wave mode is characterized primarily by zonal fluctuations. Early observations suggested these stratospheric waves behave as dry (or “free”) modes, with their energy likely originating from diabatic heating associated with moist convection in the troposphere (Holton, 1972, 1973). The theory of equatorially trapped waves and its connection to observed propagating systems of organized tropical convection and the equatorial wave modes in the lower stratosphere have been extensively studied, yet remain incompletely understood (e.g., Bergman & Salby, 1994; Chang, 1976; Chen et al., 2025; Chien & Kim, 2024; Dias et al., 2023; Garcia & Salby, 1987; Hayashi & Golder, 1994, 1997; Holton, 1972; Lindzen, 1974; Rios-Berrios et al., 2023; Salby & Garcia, 1987). In particular, uncertainties persist regarding wave attribution, amplitude maintenance, predictive capability, and the representation of convection-wave coupling in models. Consequently, substantial research effort continues to focus on understanding how tropical convection is organized across dominant temporal frequencies and zonal wavenumbers. Recent studies have also highlighted the importance of CCEWs in organizing tropical convection and modulating rainfall variability, as well as the challenges in representing them realistically in numerical models (Karthek et al., 2024; Lee et al., 2025; Roundy et al., 2023; Yamazaki et al., 2024).

CCEWs exhibit distinct spectral and propagation characteristics. These waves are not only critical

modulators of synoptic- and intraseasonal-scale convective variability (Kiladis et al., 2009), but also play a vital role in the development and evolution of large-scale organized modes of tropical variability such as the Madden–Julian Oscillation (MJO) (Roundy, 2008), monsoon intraseasonal oscillations (Goswami et al. 2006), tropical cyclogenesis (Emmanuel et al., 2021; Frank & Roundy, 2006; Schreck & Molinari, 2011) and extreme rainfall events (Mukhopadhyay et al., 2021). Improving the prediction of CCEWs are essential for improving short- to medium-range weather forecasts and intraseasonal variability. Moreover, their accurate representation in the model, particularly of spectral characteristics, phase speed, and convective coupling serves as a benchmark for assessing the fidelity of regional and global climate models, especially during the Indian Summer Monsoon (JJAS), where convective activity is highly responsive to wave dynamics. An improved representation of CCEWs in models not only enhances our understanding of tropical dynamics but also contributes to more reliable predictions of rainfall, storm development, and intraseasonal oscillations over the tropics in general and monsoon region in particular. Previous studies have documented significant deficiencies in simulating high-frequency intraseasonal variability including weak or incoherent representations of Boreal Summer Intraseasonal Oscillation (BSISO), Kelvin waves, and CCEWs in many global models, including Climate Forecast System version 2 (CFSv2) (Abhik et al., 2016; Goswami et al., 2017; Sperber et al., 2013; Waliser et al., 2009). The underrepresentation of Kelvin wave initiation and eastward propagation critically impairs the model’s ability to capture organized convection associated with the BSISO, resulting in a persistent dry bias over the Indian Ocean and Asian monsoon regions (Jiang et al., 2015; Kiladis et al., 2009; Straub & Kiladis, 2003). These studies further demonstrate that enhancing physical parameterizations and resolution significantly improves both synoptic and subseasonal variability. Despite the recognized importance of CCEWs in tropical variability and monsoon modulation, their representation and forecast evolution in operational prediction systems remains elusive. In this study, we evaluate the ability of the IITM-IMD GFS T1534

forecast system to represent key CCEW modes, including Kelvin, ER, MRG waves, and the MJO, during the boreal summer season (JJAS). Using space–time spectral diagnostics and composite analyses of OLR variability, we examine how these wave modes are represented in the model and how their characteristics evolve with forecast lead time (Day-1, Day-3, and Day-5). This analysis provides insight into the strengths and limitations of the model in capturing tropical wave variability that influences intraseasonal variability and monsoon circulation. Understanding these aspects is important for improving the prediction of tropical disturbances and their associated large-scale impacts in operational forecasting systems. The remainder of the paper is organized as follows: Sect. 2 describes the model, data, and methodology used in this study. Section 3 presents the results and discussion, and Sect. 4 summarizes the main findings.

2. Model Details, Data & Methodology

In this study, we utilize operational reforecasts generated using the operational IITM-IMD GFS model (Johnny & Prasad, 2020; Mukhopadhyay et al., 2019) at T1534 resolution. This study utilizes the reforecast data spanning the period 2000–2018. The study period (2000–2018) was selected based on the availability of a continuous reforecast dataset for the analysis. The model used is NCEP’s version 14.1.0 of Global Forecast System (GFS), and reforecasts are initialized for the 00 UTC of each day and run for 10 days forecast. The IITM GFS employs a two-time-level semi-implicit semi-Lagrangian (SL) dynamical core (Sela, 2010). Semi-Lagrangian advection and physical processes are computed on a linear, reduced Gaussian grid. The model uses a horizontal resolution of 3072×1536 grid points (~ 12.5 km) and 64 vertical hybrid sigma-pressure levels. The model GFS T1534 was operationalized for IMD under the National Monsoon Mission of Ministry of Earth Sciences (MoES), Government of India.

The model uses Real-Time Global (RTG) sea surface temperature (SST), daily sea ice analysis, and blended snow analysis as lower boundary conditions over the ocean. The planetary boundary layer (PBL)

is represented using a hybrid Eddy-Diffusivity Mass-Flux (EDMF) vertical turbulent mixing scheme, which applies mass-flux formulation under strongly unstable conditions and counter-gradient eddy-diffusivity under weakly unstable conditions (Han & Pan, 2011; Han et al., 2017). Deep convection is parameterized using the revised simplified Arakawa-Schubert (SAS) deep convection scheme (Han & Pan, 2011; Han et al., 2017; Pan & Wu, 1995), while shallow convection follows a mass-flux-based SAS scheme with scale-aware modification by Han and Pan (2011); Arakawa and Wu (2013). Cloud microphysics processes are based on the schemes of Sundqvist et al. (1989) and Zhao and Carr (1997), representing grid-scale condensation and precipitation. For radiative processes, the model uses the Rapid Radiative Transfer Model (RRTM) for both longwave and shortwave radiation (Clough et al., 2005; Iacono et al., 2000), incorporating the Monte Carlo Independent Column Approximation (McICA) to address sub-grid cloud variability, particularly in multi-layered cloud structures. Comprehensive details of the model physics can be found in Johnny and Prasad (2020); Ganai et al. (2026). The initial condition of the IITM-IMD GFS model run is generated by National Centre for Medium Range Weather Forecasting (NCMRWF) through ensemble Kalman filter (EnKF) component of hybrid global data assimilation system (GDAS) (Prasad et al., 2016).

The National Aeronautics and Space Administration (NASA) Integrated Multi-satellite Retrievals for GPM (IMERG) Final Run Version 06B daily precipitation data (Huffman et al., 2019) for the boreal summer months (June to September; JJAS) over the period 2000–2018 is employed for validating the model. This dataset offers gridded precipitation estimates at a spatial resolution of $0.1^\circ \times 0.1^\circ$ across the global tropics, making it well-suited for assessing daily rainfall characteristics and variability. In addition, the Outgoing Longwave Radiation (OLR) data from the Advanced Very High Resolution Radiometer (AVHRR) on board NOAA polar-orbiting satellites is utilized for the same period (2000–2018). This daily averaged dataset is processed on to a $2.5^\circ \times 2.5^\circ$ latitude–longitude grid with missing

values filled by interpolation (Liebmann & Smith, 1996).

For both the observational datasets and the model reforecast dataset, daily OLR anomalies are computed by removing the climatological annual cycle, which is defined as the sum of the annual mean and the first three harmonics of the seasonal cycle for each individual year. The daily anomalies are filtered in wavenumber-frequency space to isolate signals associated with CCEWs, including the Kelvin wave, the first symmetric westward-propagating equatorial Rossby wave (ER, $n = 1$), the MJO and anti-symmetric MRG waves, following the method of Wheeler and Kiladis (1999). Each filtered wave mode exhibits either symmetric or anti-symmetric structure about the equator. In this study, the Kelvin wave filter is designed to retain eastward-propagating signals with periods ranging from 2.5 to 17 days and zonal wavenumbers 1 to 14, bounded by theoretical dispersion curves corresponding to equivalent depths between 8 and 90 m. The $n = 1$ ER wave filter captures westward-propagating features with periods of 10 to 48 days and zonal wavenumbers 1 to 10, also bounded by ER dispersion curves for equivalent depths of 8 to 90 m. To isolate the MJO, a separate filter is applied to retain eastward-propagating disturbances with periods of 30 to 60 days and zonal wavenumbers 1 to 5, which captures the dominant intraseasonal variability over the tropics associated with the MJO. The MRG wave filter retains westward propagating antisymmetric disturbances with periods ranging from 3 to 6 days and zonal wave numbers 1 to 10 (Fig. 8a & e).

3. Results and Discussion

3.1. Global Outgoing Longwave Radiation (OLR) Distribution

Globally, OLR serves as a key indicator of deep convection and cloud-top temperatures, where lower OLR values generally correspond to enhanced convective activity and high, cold cloud tops (Murakami, 1980). It is extensively used in climate and weather studies to monitor tropical convection, detect equatorial wave activity, and understand large-scale

energy balance (Liebmann & Smith, 1996; Trenberth et al., 2009; Wheeler & Kiladis, 1999). The global distribution of OLR exhibits strong spatial and temporal variability, closely linked to the climatology of convection, surface temperature, and cloud cover. In order to validate the model performance in capturing the convection, the NOAA OLR mean state is compared with the OLR from GFS T1534 model for Day-1, Day-3 and Day-5 lead forecasts during summer monsoon season (JJAS) for the period 2000–2018 (Fig. 1). Observations from NOAA indicate that the equatorial Indian Ocean, Maritime Continent, Western Pacific, and parts of Central/South America are characterized by persistently low OLR values ($180\text{--}220\text{ W m}^{-2}$), reflecting strong convective activity associated with the monsoon system (Intertropical Convergence Zone (ITCZ), monsoon convection). The subtropical oceans, eastern Pacific and Africa (north of the equator) shows high OLR values ($> 260\text{ W m}^{-2}$) as these are the regions of subsidence/dry environment with clear skies. Over the global tropics, this pattern is generally captured by the model forecasts on days 1, 3, and 5, although a slight overestimation of convection becomes apparent as the forecast lead time increases.

The biases in the seasonal mean simulations (days 1, 3 and 5) relative to observations, are shown in the Fig. 2. The model shows a negative bias with low OLR values (-5 to -15 W m^{-2}) i.e., overestimation of convection over the Western Indian ocean, Bay of Bengal, Southeast Asia, tropical south America and central Africa which are the main tropical convective zones. The model shows a positive bias (around $10\text{--}20\text{ W m}^{-2}$) (underestimation of convection) over the eastern Indian Ocean, Maritime continent, and equatorial Pacific. These biases increase consistently as the forecast lead time increases. As OLR is closely tied to tropical convection, and rainfall, these biases suggest limitations in the model's representation of convection. Such errors may influence monsoon prediction, MJO and equatorial wave simulations, as well as associated climate feedbacks. The positive bias over the head Bay of Bengal at all lead times suggests limitations in the model's representation of monsoon convection over this region, which serves as a primary breeding ground for monsoon lows and depressions.

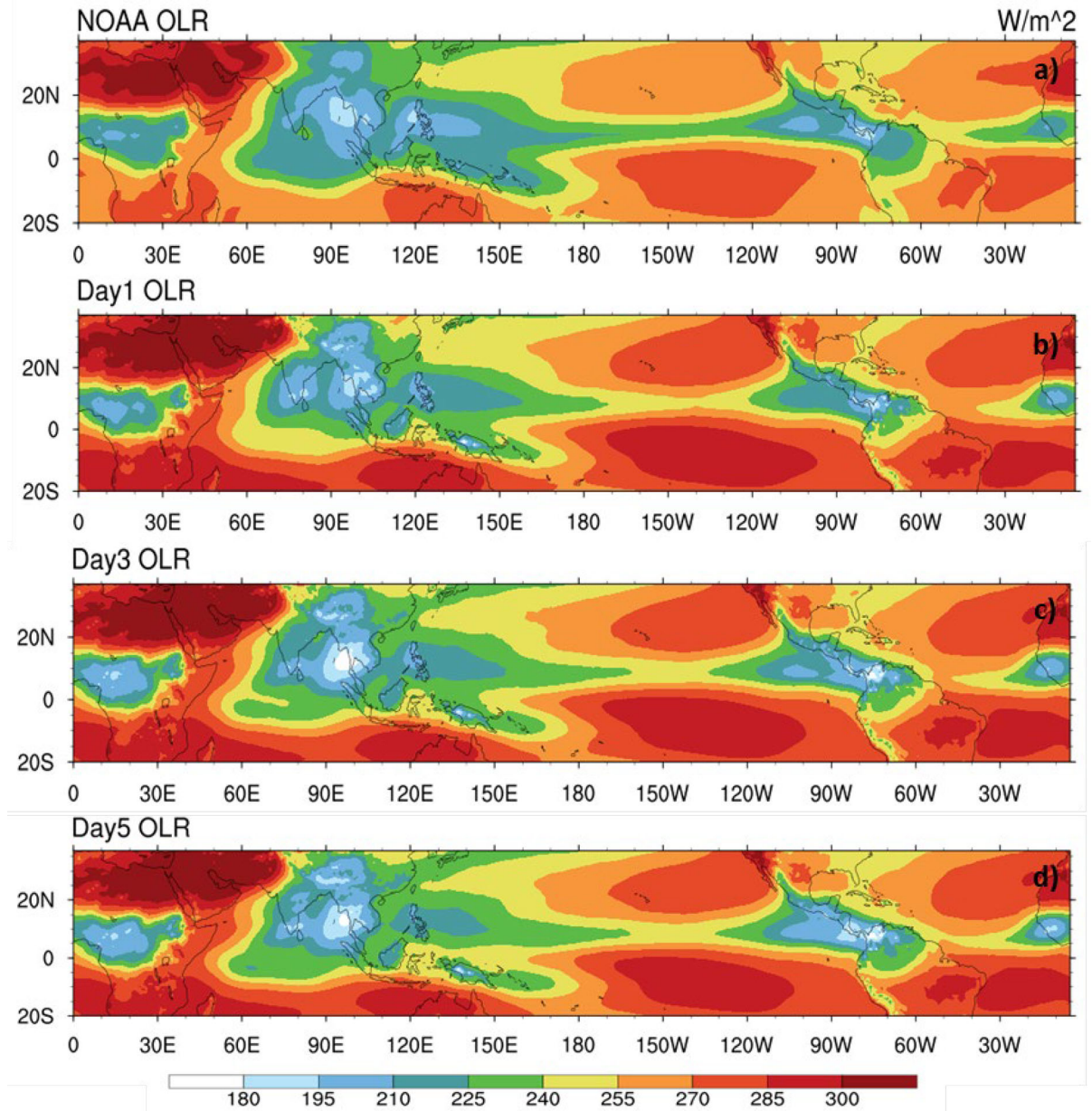


Figure. 1

OLR (W m^{-2}) mean state during JJAS 2000-2018 **a** NOAA observations, **b** Day 1, **c** Day 3, and **d** Day 5 forecast from the GFST1534 model

3.2. Mean Precipitation Distribution

The tropical regions (approximately 15°S-15°N) exhibit the highest annual rainfall, particularly along the ITCZ, where converging trade winds and vigorous convection promote persistent precipitation. Figure 3 presents the seasonal mean precipitation

(mm day^{-1}) during the analysis period with the top panel showing the IMERG precipitation and subsequent panels illustrating days 1, 3 and 5 forecasts. In the IMERG observations, rainfall is concentrated in the tropical belt with pronounced maxima over the Western Ghats, Eastern shore of the Bay of Bengal, the maritime continent, Western Pacific and Eastern

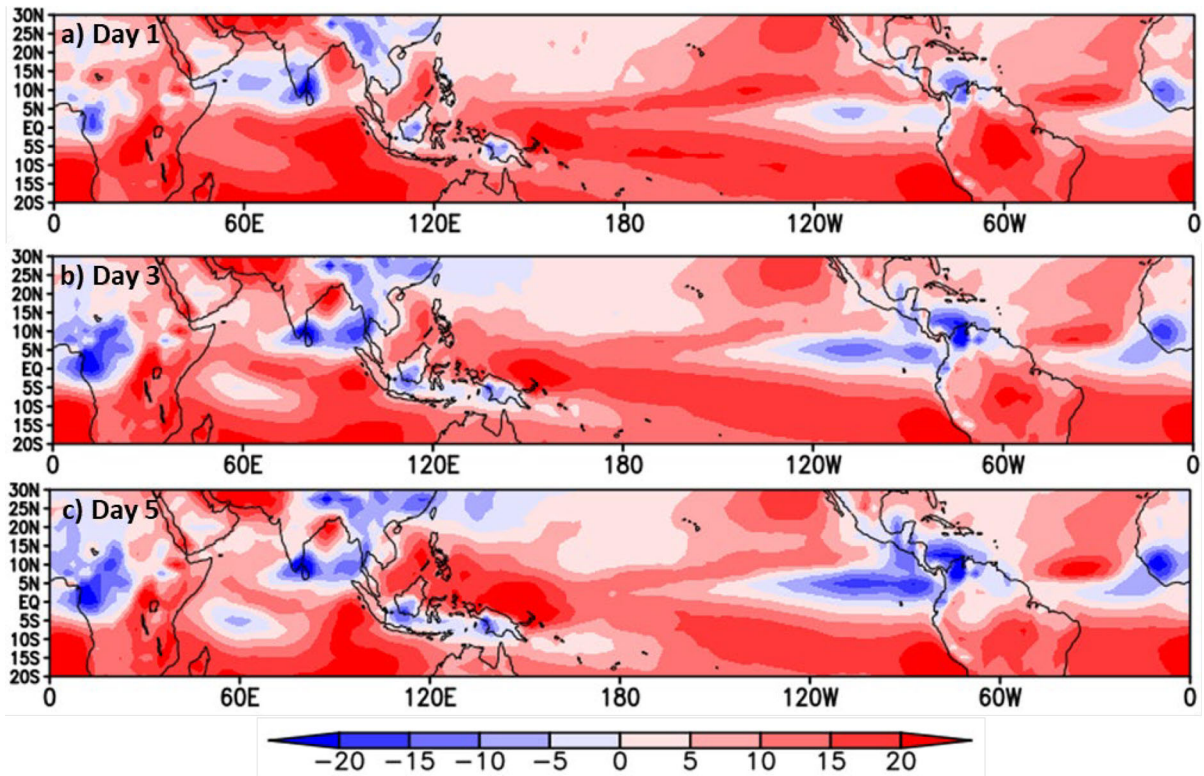


Figure. 2

OLR (W m^{-2}) bias (model-observation) during JJAS 2000-2018 for **a** Day 1, **b** Day 3, and **c** Day 5 forecast from the GFST1534 model

Equatorial Indian Ocean (EEIO). These regions are governed by persistent deep convection associated with the ITCZ and active monsoon systems. Based on long-term IMERG data (Huffman et al., 2019) the global rainfall distribution demonstrates substantial spatial variability, closely linked to the large-scale circulation and convective processes (e.g., Tan et al., 2019). The model reasonably captures the location of major rainfall zones, although it increasingly overestimates rainfall amounts with forecast lead time. Moreover, the model exhibits a tendency toward a southward extension of precipitation over the central and eastern Pacific with increasing lead time. While day 1 forecasts depict a relatively equatorially confined precipitation band, by day 3 a weak secondary maximum begins to emerge south of the equator, and by day 5 enhanced rainfall is evident near 5°S-10°S over parts of the central Pacific. This feature is absent in the IMERG observations, indicating a developing southward precipitation bias in

the model forecasts. Such biases in tropical rainfall distribution are consistent with long-standing challenges in global models (Mechoso et al., 1995; Lin 2006; Zhang & Wang, 2006; Li & Xie, 2014; Oueslati & Bellon, 2015).

Figure 4 illustrates the rainfall biases of the model (day 1, day 3 and day 5) relative to the observations. The model shows positive bias with high precipitation values ($3\text{--}5 \text{ mm day}^{-1}$) (overestimation of convection/rainfall) over Western Ghats, Bay of Bengal, Maritime continent, central and eastern India and shows negative bias of -3 to -5 mm day^{-1} (underestimation of rainfall) over equatorial Indian ocean and Western Pacific. These biases indicate regional deficiencies in the model's representation of convection, orography, and large-scale circulation, particularly within tropical convective zones. Consistent with the positive OLR bias over the head Bay of Bengal—indicative of weaker convection—the model also shows a negative rainfall bias over the

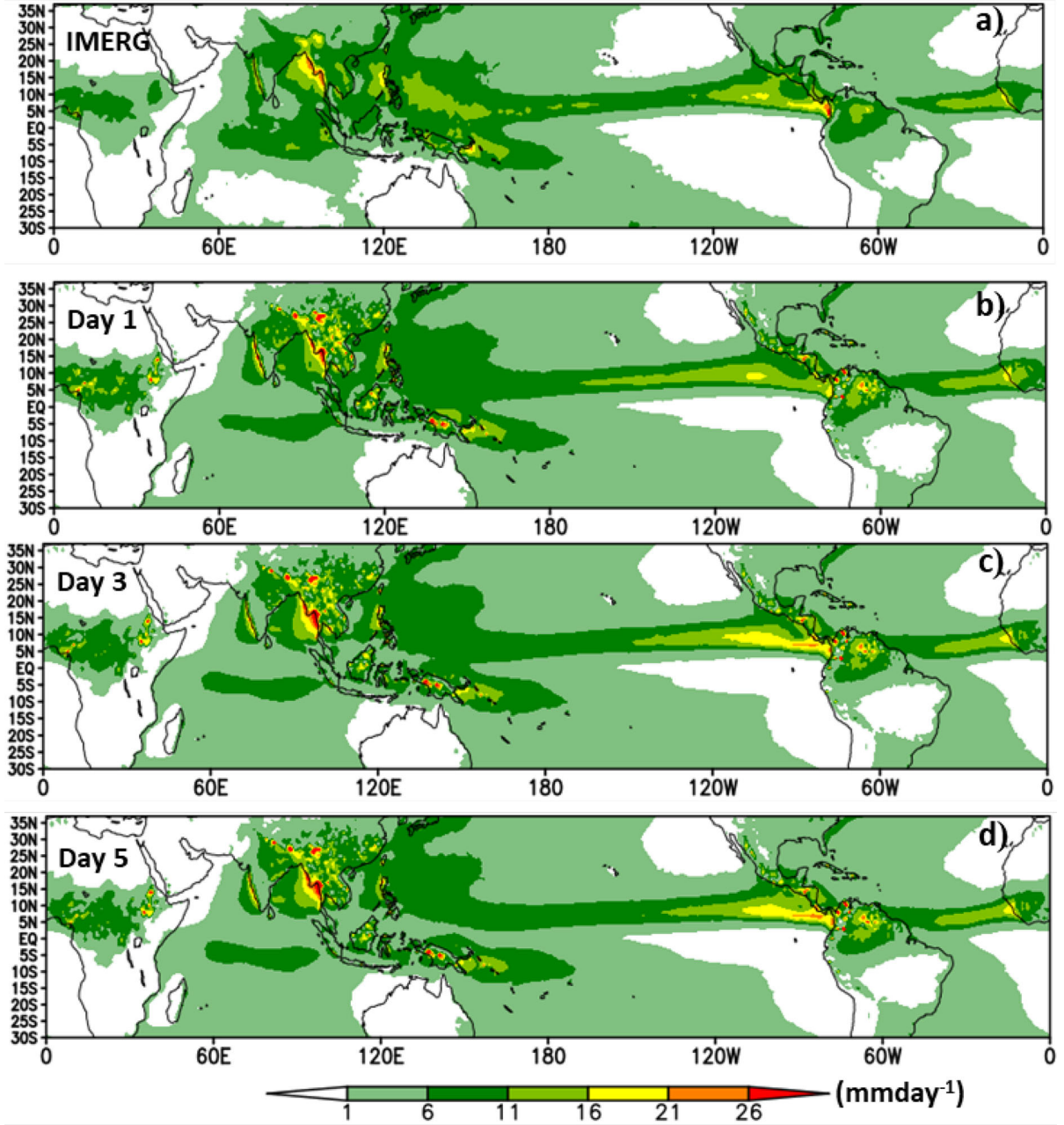


Figure. 3

Seasonal precipitation (mm day⁻¹) during JJAS 2000-2018 **a** IMERG, **b** Day 1, **c** Day 3, and **d** Day 5 forecast from the GFST1534 model

same region (Fig. 4), suggesting a coherent underestimation of convective activity over the region. Moreover, the overestimation (underestimation) to the south (north) of equator in the Western Pacific may influence the distribution of diabatic heating

through latent heat release and its associated circulation response, which could contribute to the model forecast bias (Gill, 1980).

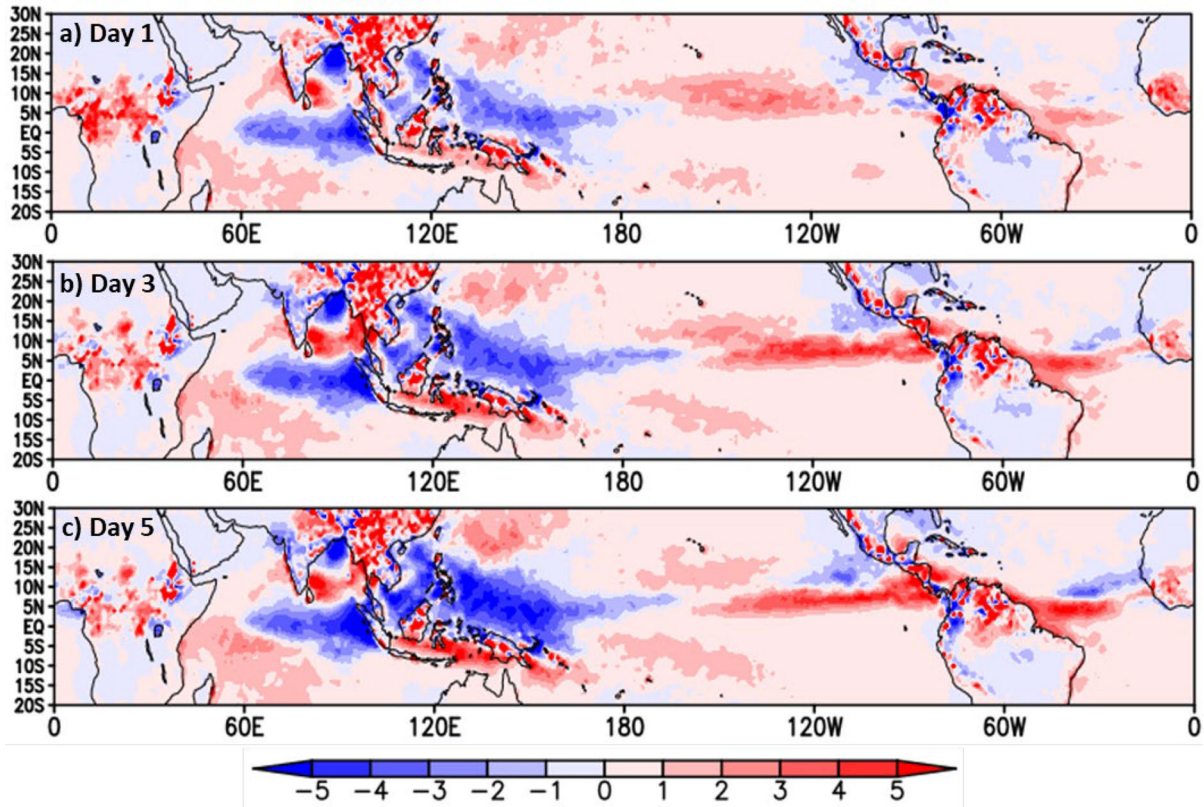


Figure. 4

Seasonal precipitation (mm day^{-1}) bias during JJAS 2000–2018 **a** Day 1, **b** Day 3, and **c** Day 5 forecast from the GFST1534 model

3.3. Precipitation Probability Distribution Function (PDF) Distribution

To further investigate the rainfall bias, the probability distribution function (PDF) of daily rainfall during the JJAS season (2000–2018) is shown in Fig. 5. Figure 5a presents the PDF of daily rainfall over the global tropics (20°S – 20°N), derived from IMERG observations and GFS T1534 model forecasts at lead times of Day-1, Day-3, and Day-5. The IMERG dataset exhibits a peak in moderate rainfall events, particularly in the 10 – 30 mm day^{-1} range, reflecting the typical distribution of tropical convection. In contrast, the model forecasts overestimate both light rainfall (0 – 10 mm day^{-1}) and moderate rainfall (10 – 30 mm day^{-1}) while underestimating heavy rainfall category ($> 50 \text{ mm day}^{-1}$), especially the extreme tail ($> 150 \text{ mm day}^{-1}$). These biases become more pronounced with increasing lead time. The overestimation of lighter rainfall category in

operational GFS has also been reported in earlier studies (Ganai et al., 2021; Mukhopadhyay et al., 2019). This progressive divergence between model and observations indicates a degradation in rainfall simulation with forecast lead time. Such biases, particularly the overestimation of moderate rainfall, may be associated with limitations in the representation of convection and moist processes in numerical models (Dai, 2006; Mehran et al., 2014; Stephens et al., 2010). These systematic errors can influence the representation of the intensity and frequency of precipitation events in the tropics.

A similar analysis is performed over Central India (CI) (16.5° – 26.5°N , 74.5° – 86.5°E), one of the major precipitation receiving regions during the summer monsoon season, to evaluate the model's forecast skill in simulating regional rainfall characteristics. The Fig. 5b shows the PDF of daily rainfall over CI, comparing IMERG observations with GFS T1534

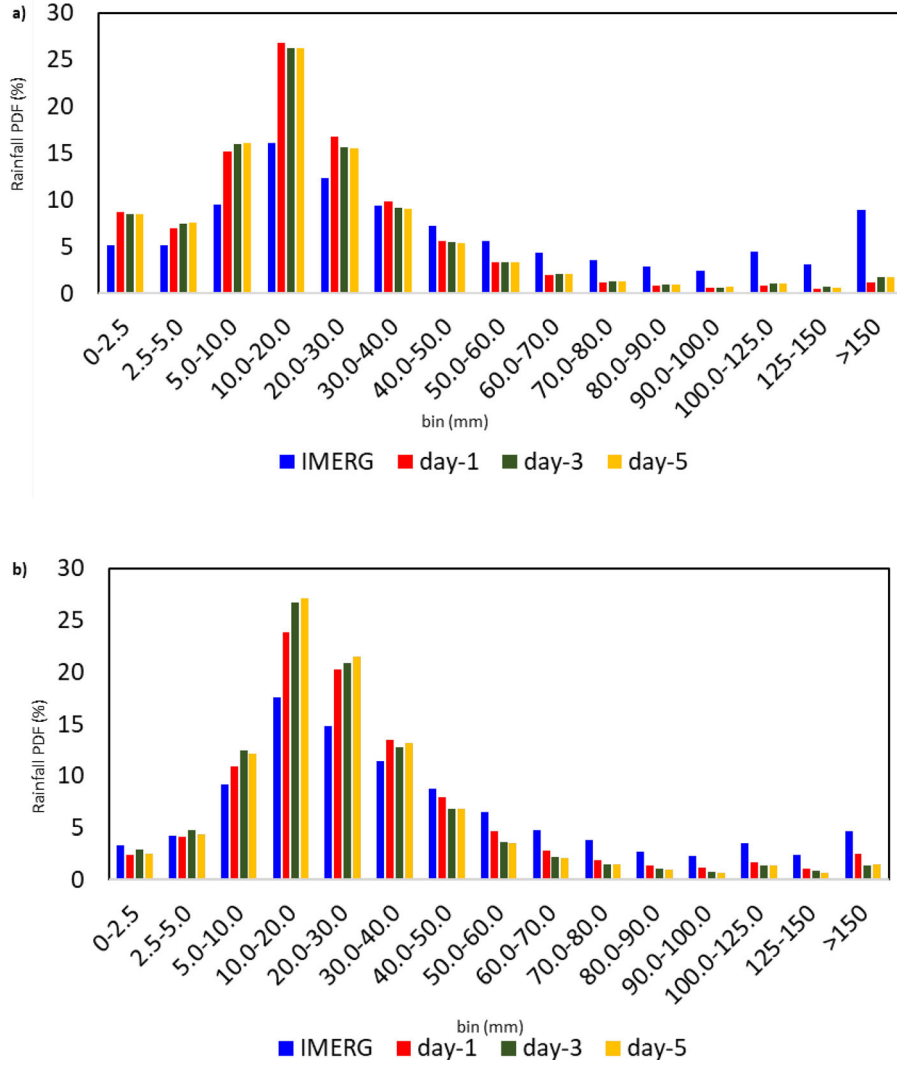


Figure. 5

The rainfall PDF (%) vs. rain rate (mm day^{-1}) categories during JJAS 2000-2018 over the **a** Global tropics and **b** central India for Day 1, Day 3, and Day 5 forecast lead times from the model and compared with IMERG data

model forecasts at Day-1, Day-3, and Day-5 lead times. IMERG shows a peak in moderate rainfall intensities ($10\text{--}30 \text{ mm day}^{-1}$), consistent with monsoon convection over the region. The model forecasts capture the overall distribution shape but tend to overestimate the frequency of light ($2.5\text{--}10 \text{ mm day}^{-1}$) and moderate rainfall ($10\text{--}30 \text{ mm day}^{-1}$), particularly evident at Day-3 and Day-5, consistent with the model's known wet bias. This is in line with previous findings that GFS-based models tend to overproduce weak-to-moderate

rainfall over Central India (Mukhopadhyay et al., 2019; Sridevi et al. 2022; Ganai et al., 2024). Conversely, the model tends to underrepresent the occurrence of high-intensity rainfall ($> 50 \text{ mm day}^{-1}$), particularly in the extreme tail ($> 150 \text{ mm day}^{-1}$), which IMERG captures more frequently. These biases may influence the simulation of monsoon extremes and rainfall distribution patterns over Central India, a region highly sensitive to precipitation variability during the summer monsoon season (Prakash et al. 2021).

It is well established that South Asian summer monsoon rainfall is strongly modulated by the intraseasonal variability including the quasi-stationary 30–40 day oscillation first identified by Yasunari (1980), the BSISO (Goswami & Mohan, 2001; Goswami et al., 2006), and broader global tropical modes such as the MJO. These signals all involve CCEWs, including Kelvin, MRG, and ER waves, which play a key role in organizing rainfall across the Indian Ocean and global tropics (Wheeler & Kiladis 1999; Kiladis et al. 2009; Roman-Stork & Bourassa 2022). So, filtering OLR, rainfall and wind data is essential to isolate BSISO/MJO and associated CCEWs signals. These oscillations/modes are further analysed/studied in the next sub sections.

3.4. Percentage of Daily Variance by 20–100 Day and 2–20 Day Mode

Figure 6 presents the percentage of daily OLR variance explained by the 20–100 day intraseasonal mode during the boreal summer (JJAS) over the global tropics for the period 2000–2018, based on NOAA observations and GFS T1534 model forecasts at Day-1, Day-3, and Day-5 for the period 2000–2018. The shaded map shows the ratio of bandpass-filtered variance (20–100 day intraseasonal variance) to the unfiltered variance at each grid point. Contour lines overlay the filtered variance, in the units (W m^{-2})². By comparing both ratio and filtered variance variability, the figure allows better identification of regions where intraseasonal modes (e.g., MJO, BSISO) strongly influence daily changes, versus areas dominated by other variability. The observations reveal strong intraseasonal variability in key convective regions such as the Arabian Sea, Western India, and the Western Pacific, where the OLR variance contribution from 20–100 day oscillations exceeds 35–40%. These regions are strongly influenced by phenomena such as the MJO and BSISO, which modulate deep convection on intraseasonal timescales (Abhik et al., 2016; Lee et al., 2015; Waliser et al., 2009).

The GFS T1534 model captures the broad spatial distribution of intraseasonal variance reasonably well, but the magnitude and structure of the signal degrade with increasing lead time. At Day-1, the

model underestimates the variance in the equatorial Pacific and parts of the Indian Ocean, though it maintains a fair representation over the Bay of Bengal. By Day-3, a slight improvement is evident over the northern Indian Ocean, eastern Pacific, but the signal becomes more diffuse over the Maritime Continent and Western Pacific. At Day-5, the model shows continued attenuation of the intraseasonal signal, particularly over the Indian subcontinent and equatorial Indian Ocean, suggesting reduced skill in maintaining coherent large-scale variability. This behaviour is consistent with known model limitations in representing moist convection, cloud-radiation interactions, and coupling with surface processes, all of which are crucial for capturing intraseasonal variability (Abhik et al., 2016; Sperber et al., 2013).

Similarly, 2–20 day mode of high frequency variability is also compared between the model and observation (Fig. 7). The shaded map shows the ratio of bandpass-filtered variance (2–20 day variance) to the unfiltered variance at each grid point. Contour lines overlay the filtered variance, in the units (W m^{-2})². Compared to the low frequency intraseasonal variability, high frequency variance is notably high in the NOAA observations. In contrast, the GFS model forecasts exhibit a noticeable underestimation of the 2–20 day OLR variance across most tropical regions, with the suppression becoming more pronounced at longer lead times (from Day-1 to Day-5). The model shows limited ability to capture the observed intensity and spatial distribution of high-frequency convective variability. This underestimation suggests limitations in representing convectively coupled modes and sub-synoptic disturbances, which may be related to challenges in simulating convection and its interaction with large scale waves in the model (Abhik et al., 2016; Sperber et al., 2013; Waliser et al., 2009; Wolding et al., 2025). To further examine the model bias, we analyse and validate the model's CCEWs against observations in the following sections.

3.5. Wavenumber-Frequency Spectrum of OLR

Space-time spectral analysis has become a standard metric to quantify properties of equatorial waves in the observations (Kiladis et al., 2009 and

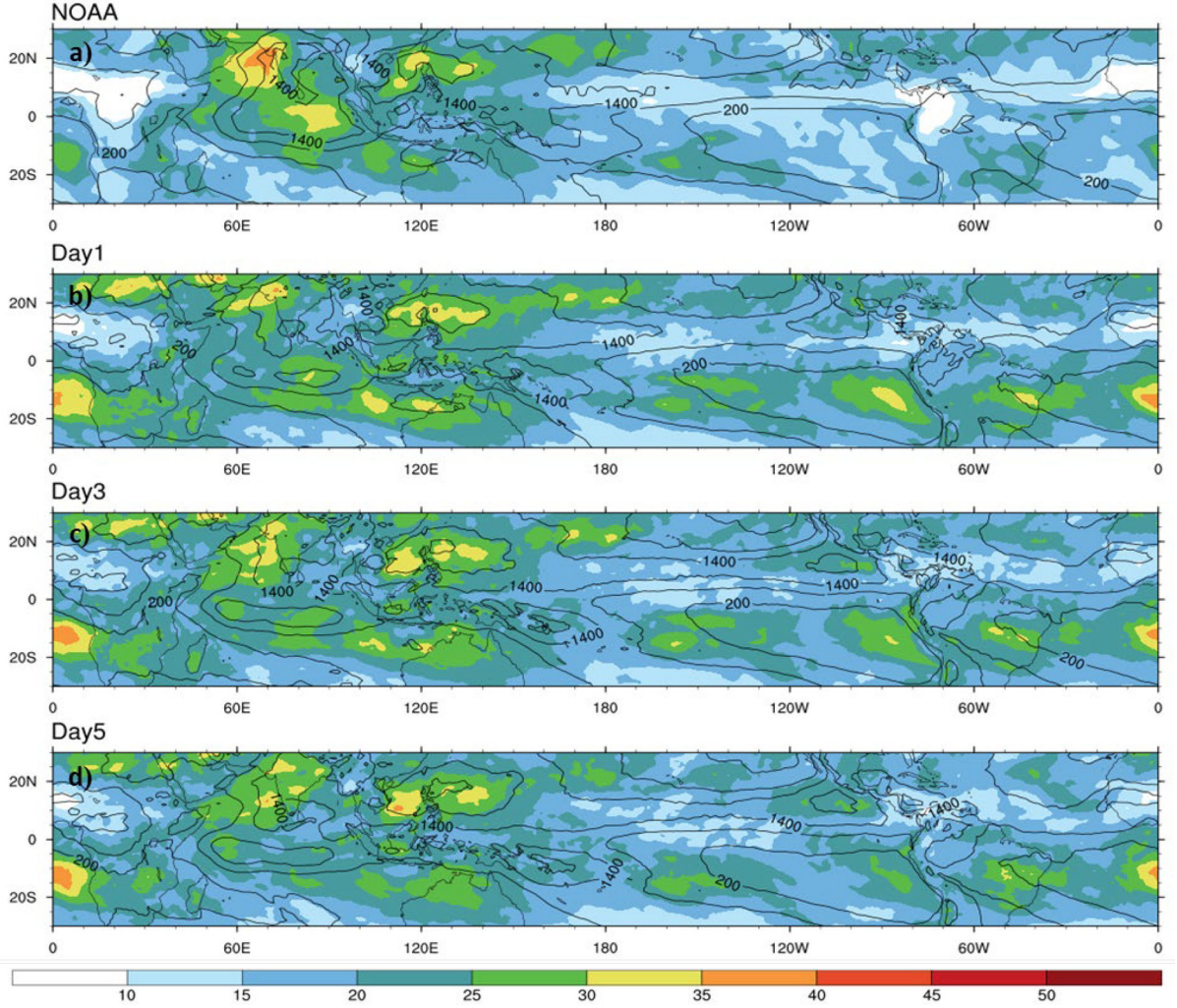


Figure. 6

Percentage of total daily OLR variance (W m^{-2})² explained by the 20-100 day mode during JJAS (2000-2018) based on- **a** NOAA Observations, **b-d** Day 1, Day 3, and Day 5 model reforecast respectively. The colour-shaded map shows the ratio of bandpass-filtered variance to total (unfiltered) variance at each grid point. Contours represent the filtered variance (W m^{-2})²

references there in) and models (Abhik et al., 2016; Guo et al., 2015; Lin et al., 2006; Straub et al., 2010). The space-time spectral analysis technique, as introduced by Wheeler and Kiladis (1999), is applied to equatorially averaged OLR anomalies at latitudinal range of 15°S-15°N. This method decomposes the OLR anomalies into their frequency and zonal wavenumber components, enabling the identification of dominant equatorial wave modes such as Kelvin waves, ER waves, MRG waves, and the Madden-Julian Oscillation (MJO). Figure 8 (a-d) presents

the space-time spectra for the equatorially symmetric component of OLR anomalies. The observed (NOAA) spectrum exhibits a pronounced regions of enhanced spectral power in both zonal wavenumber and frequency, closely tracing the theoretical dispersion curves of CCEWs. Notably, Kelvin waves are evident with periods ranging from approximately 2.5 to 17 days, ER waves between 10 and 48 days, and the MJO showing peak power in the 30–60 day range. In the model, the spectral peaks corresponding to the MJO and Kelvin waves are notably weaker,

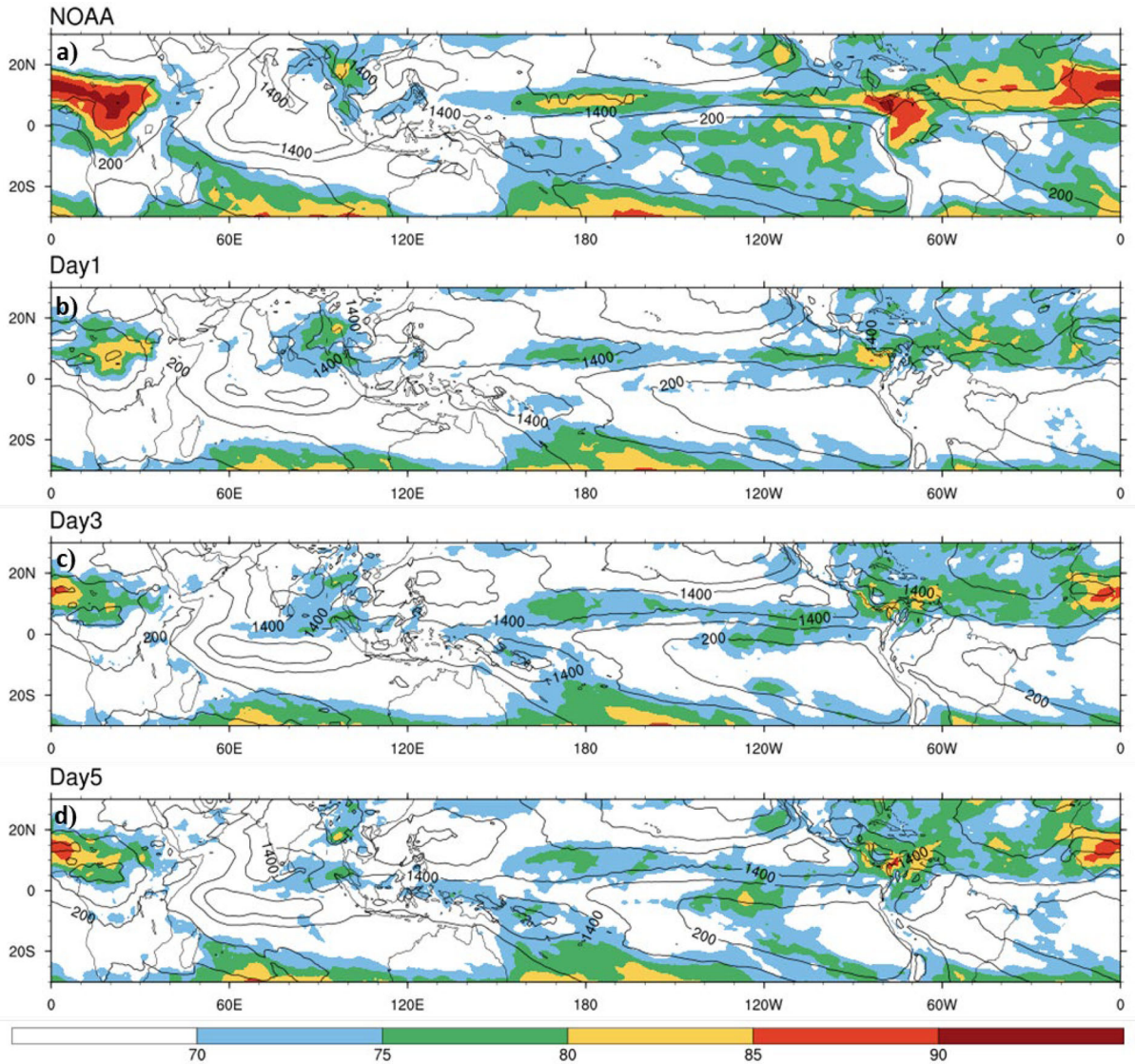


Figure. 7

Percentage of total daily OLR variance ($\text{W m}^{-2,2}$) explained by the 2-20 day mode during JJAS (2000-2018) based on- **a** NOAA Observations, **b-d** Day 1, Day 3, and Day 5 model reforecast respectively. The colour-shaded map shows the ratio of bandpass-filtered variance to total (unfiltered) variance at each grid point. Contours represent the filtered variance (W m^{-2})²

while the first-mode ER ($n = 1$) wave appears stronger than observed (Fig. 8b). Despite these differences, the model forecasts for days 1, 3, and 5 reasonably reproduce the overall spectral structure observed in the analysis. In addition to the symmetric component, the anti-symmetric component of OLR anomalies (Fig. 8 e—h) were analysed to examine equatorially anti-symmetric wave modes. These spectra primarily highlight variability associated with

MRG waves. The observed spectra exhibit enhanced power within the theoretical MRG dispersion curves, typically occurring at periods of approximately 3–6 days and westward zonal wavenumbers. The model forecasts reproduce a weak but identifiable signal within the MRG wave band at shorter lead times. However, similar to the behaviour noted for Kelvin, ER, and MJO modes in the symmetric spectra, the spectral power associated with MRG

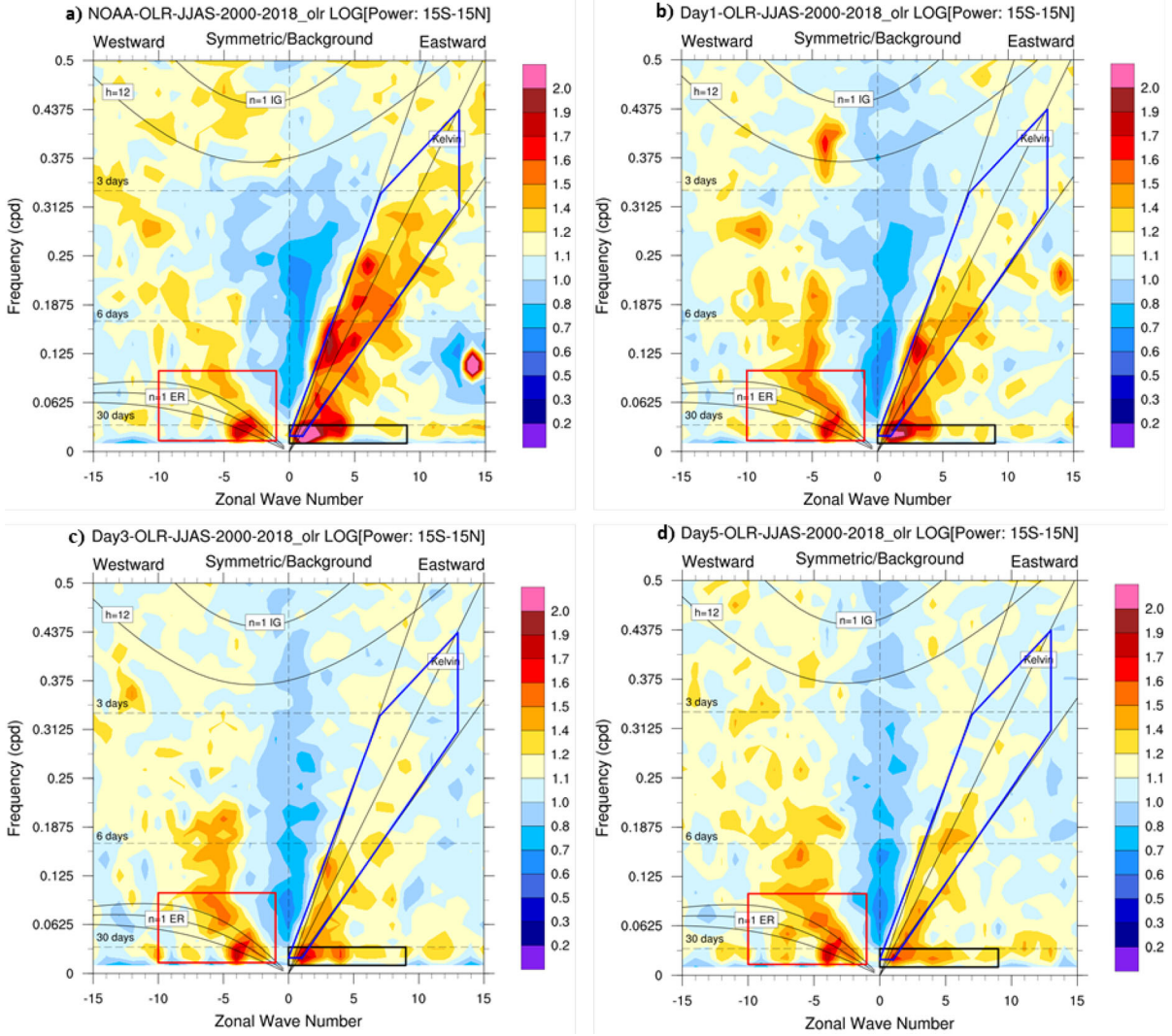
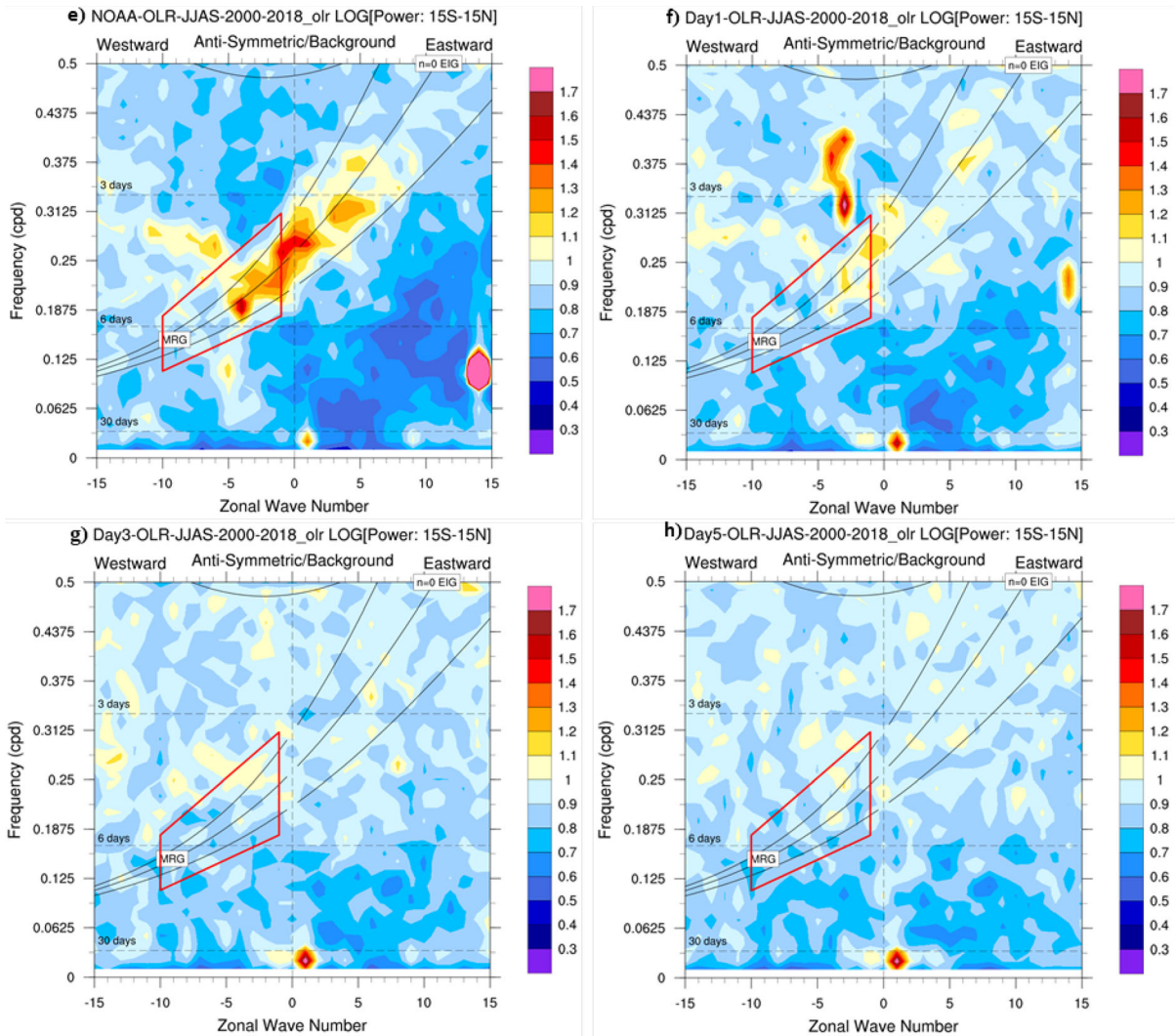


Figure 8

Wavenumber - frequency distribution of spectral-power divided by estimate background spectra for equatorially symmetric and antisymmetric OLR anomalies (W m^{-2}) during 2000-2018. The power has been summed over $15^{\circ}\text{S} - 15^{\circ}\text{N}$ latitude, and the base-10 logarithm is used for taken for plotting. Theoretical shallow-water dispersion curves corresponding to equivalent depths of $h = 12, 25$, and 50 m are plotted following Wheeler and Kiladis (1999). Panels represent **a,e** observations (NOAA OLR), **b,f** Day 1, **c,g** Day 3, and **d,h** Day 5 forecasts from the model. Polygons indicated the wavenumber-frequency domains used to isolate specific equatorial modes (e.g., Kelvin, ER, MRG waves and MJO)

waves decreases with increasing forecast lead times. This indicates that the representation of these anti-symmetric disturbances becomes progressively weaker in the Day-3 and Day-5 forecasts. Furthermore, the model spectra exhibit insufficient power at higher frequencies, indicating an underrepresentation of faster-propagating wave components. This deficiency is consistent with the reduced representation

of high frequency variability (approximately 2–20 days), which includes kelvin, MRG wave activity discussed in the previous Sect. 3.4. Although both symmetric and antisymmetric wave modes are present in the tropical spectrum, this study primarily focuses Kelvin, ER, and MJO modes because they contribute most strongly to boreal summer intraseasonal variability and large-scale monsoon

Figure. 8
continued

modulation. Nevertheless, the antisymmetric component is also examined to provide a more complete representation of equatorial wave activity. The weakening of MRG wave activity at longer lead times suggests limitations in the model's representation of high-frequency monsoon transients, which are generally associated with high-frequency CCEWs and their propagation.

3.6. Spatial Distribution of CCEWs and MJO

The region of enhanced Kelvin wave variance over the equatorial Indian Ocean consistent with previous studies identifying this area as a preferred region for convectively coupled kelvin wave activity (Straub & Kiladis, 2003; Zhang, 2005). These waves modulate the timing, intensity, and organization of deep convection and contribute to the eastward propagating components of the BSISO (Jiang et al., 2015; Kiladis et al., 2009). Figure 9 shows the global distribution of Kelvin wave variance during the

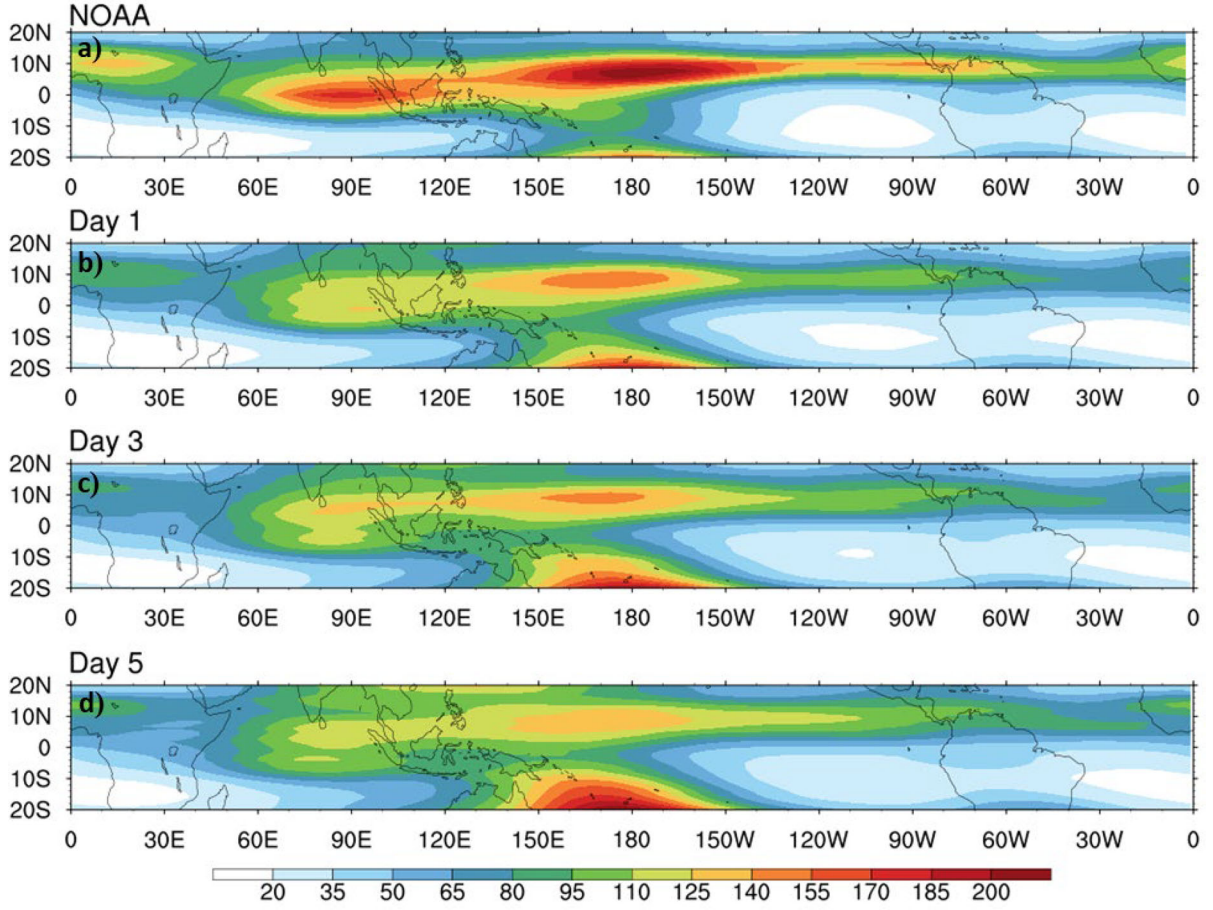


Figure. 9

Spatial distribution of OLR variance ($\text{W m}^{-2})^2$ of Kelvin waves during the period 2000-2018 for JJAS **a** NOAA OLR, **b** Day 1, **c** Day 3, and **d** Day 5 forecast lead times from the GFS T1534 model

boreal summer, illustrating the spatial structure of OLR variance associated with convectively coupled Kelvin waves (CCKWs). The analysis reveals a clear concentration of Kelvin wave activity along the equatorial belt, with pronounced variance over the central-eastern Indian Ocean, Maritime Continent, and extending into the Western-central Pacific Ocean. The strongest variance, exceeding $100 (\text{W m}^{-2})^2$, is observed over the Indian Ocean, marking it as a prominent region of Kelvin wave activity during the monsoon season and this distribution reflects the equatorially trapped nature of Kelvin waves (Fig. 9a). The observed spectral patterns confirm that Kelvin wave activity aligns closely with major monsoon convection zones during JJAS, and these waves play

a definitive role in modulating tropical intraseasonal variability, and monsoon rainfall (Roundy, 2008; Wheeler & Kiladis, 1999). The simulated CCKWs representing the similar features as observations, but the magnitude is considerably underestimated across the global tropics from day 1 to day 5 (Fig. 9 b, c & d).

Figure 10 illustrates the seasonal distribution of ER wave variance ($\text{W m}^{-2})^2$ during boreal summer. The variance is primarily confined within $\pm 10^\circ$ latitude, consistent with the equatorially trapped structure of ER waves. A pronounced maximum appears over the Western Pacific (120°E - 180°E), with a secondary enhancement over the eastern Pacific. The model reproduces this spatial structure

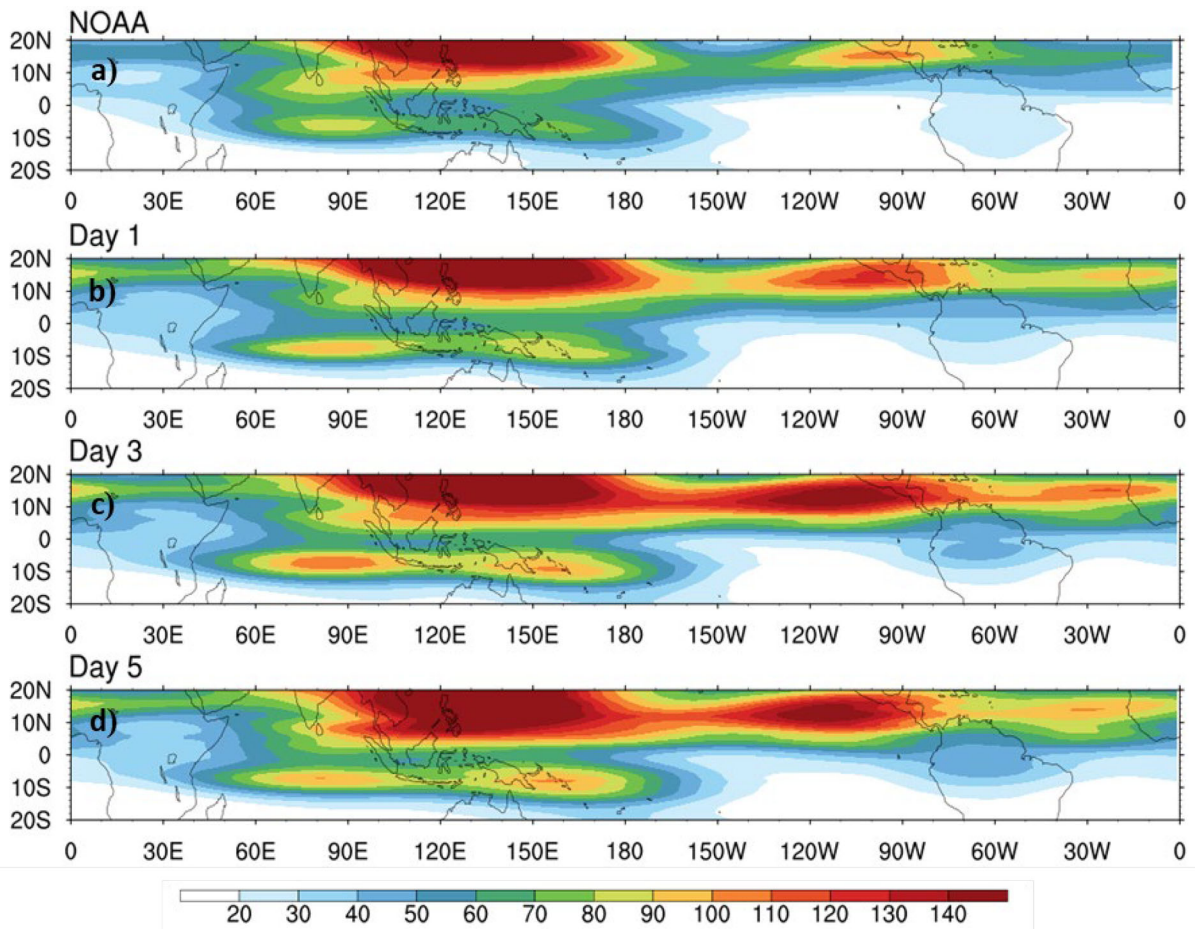


Figure. 10

Spatial distribution of OLR variance (W m^{-2})² of ER waves during the period 2000–2018 for JJAS **a** NOAA OLR, **b** Day 1, **c** Day 3, and **d** Day 5 forecast lead times from the GFS T1534 model

reasonably well, although it slightly overestimates the magnitude of variance compared to observations. The enhanced ER variance over the Western Pacific suggests active wave-convection interactions in the warm pool region during boreal summer. While ER waves contribute to large-scale circulation variability, their direct influence on convective rainfall is comparatively weaker than that of Kelvin waves, likely due to their predominantly rotational structure. The simulated variance pattern is consistent with previous studies that highlight the role of ER waves in modulating intraseasonal variability over the tropical Pacific and monsoon regions (Kiladis et al., 2009; Roundy & Frank, 2004). Unlike Kelvin waves, ER waves are associated with broader spatial scales and

contribute to the background modulation of intraseasonal variability, including BSISO-related variability and monsoon breaks (Kiladis et al., 2009; Wheeler & Kiladis, 1999).

The global distribution of MJO variance (W m^{-2})² based on the observations and model during the analysis period is shown in Fig. 11. In the observations, the strongest MJO variance is seen to be dominant over the equatorial Indian Ocean, Maritime Continent, Western Pacific, and eastern Pacific Ocean. In the model, the spatial structure appears more diffused compared to observations, and the amplitude (or sharpness) gradually decreases with increasing lead time. This smearing bias is consistent with documented limitations in GCMs, which often

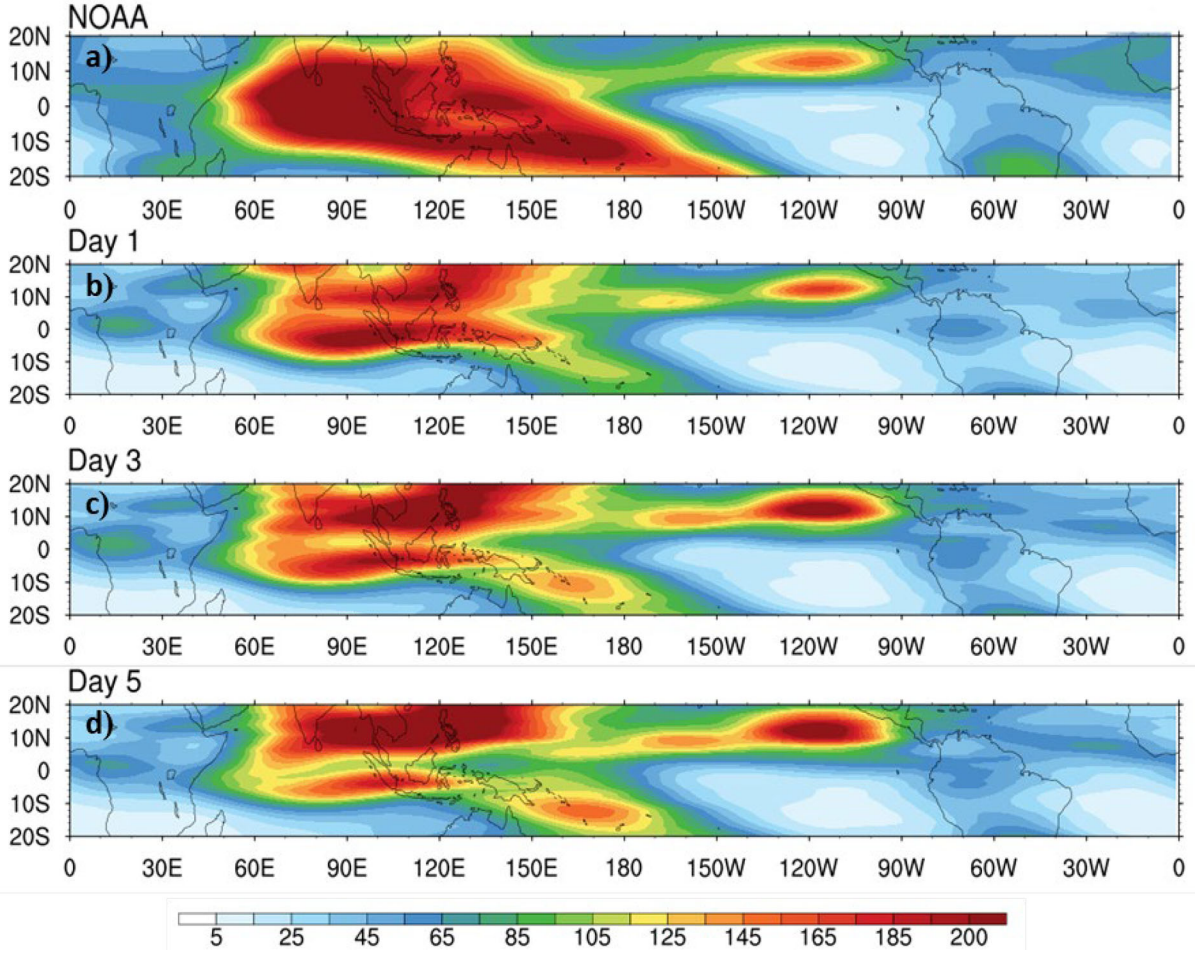


Figure. 11

Spatial distribution of OLR variance ($\text{W m}^{-2})^2$ of MJO during the period 2000-2018 for JJAS **a** NOAA OLR, **b** Day 1, **c** Day 3, and **d** Day 5 forecast lead times from the GFS T1534 model

struggle to reproduce the observed coherence, strength, and eastward propagation of MJO events particularly in Maritime continent region (Ahn et al., 2020; Le et al., 2021). While the wavenumber-frequency spectral power of the MJO (Fig. 8) typically diminishes by day 5, the spatial distribution of OLR variance (Fig. 11) at the same lead time still shows organized large scale convective patterns. This suggests that the model retains the broad spatial footprint of tropical convection even when the temporal coherence or amplitude of the MJO signal weaken. Such behaviour indicates a tendency of the model to better represent spatial structures than temporal variability of the MJO signal, consistent

with previous studies (Lin et al 2008; Waliser et al 2008; Sperber et al 2013), although many models continue to underperform in preserving MJO signal strength beyond short lead times. The core region of MJO activity lies between 10°S-10°N and 60°E-180°E, consistent with the tropical warm pool where deep convection is most persistent. The signal is equatorially confined, highlighting the MJO's nature as a convectively coupled equatorial mode.

To quantitatively assess the evolution of forecast errors with lead time, we computed the domain-averaged root mean square error (RMSE) of the filtered OLR wave components (Table 1) over the tropical belt (15°S-15°N). The RMSE values show a

Table 1

Domain-averaged root mean square error (RMSE) of filtered OLR wave components over 15°S–15°N for Day-1, Day-3, and Day-5 forecasts (units: $W\ m^{-2}$)

Wave Type	Day-1	Day-3	Day-5
Kelvin	9.67	10.59	11.11
ER	5.58	6.89	7.79
MJO	3.21	4.07	4.87

consistent increase with forecast lead time for all wave types considered. For Kelvin waves, the RMSE increases from 9.68 on Day-1 to 10.59 on Day-3 and 11.11 on Day-5. Similarly, ER waves exhibit an increase from 5.59 (Day-1) to 6.90 (Day-3) and 7.79 (Day-5). The MJO component also shows increasing errors, with RMSE values rising from 3.21 on Day-1 to 4.07 on Day-3 and 4.87 on Day-5. The increase in RMSE across all wave types indicates a gradual degradation of forecast skill with increasing lead time, consistent with the qualitative patterns observed in the spatial composites. The larger RMSE associated with Kelvin waves suggests that the model may have greater difficulty representing fast eastward-propagating convective disturbances and their associated coupling with convection. In contrast, the smaller RMSE for the MJO suggests relatively better representation of large-scale, slowly varying intraseasonal variability, which evolves on longer spatial and temporal scales.

Previous multi-model studies have documented systematic deficiencies in the representation of CCEWs in CMIP3 and CMIP5 models, including weak Kelvin wave variance and insufficient high-frequency spectral power (Hung et al., 2013; Lin et al., 2006). Although some improvements have been reported in CMIP6 models, biases in the amplitude and spectral distribution of tropical waves persist, particularly at synoptic and intraseasonal timescales (Ahn et al., 2020; Ji et al., 2025; Jiang et al., 2020). Since the IITM-IMD GFS serves as an operational forecasting system over the Indian region, it is important to assess its ability to represent CCEWs is important. The magnitude and spatial distribution of CCEW variance simulated by the IITM-IMD GFS are broadly consistent with these

findings, including underrepresentation of high-frequency variability and a modest overestimation of ER variance. This suggests that such biases may reflect persistent challenges in representing convection-wave coupling across both climate and forecast models.

4. Summary and Conclusions

This study evaluates the fidelity of the IITM-IMD GFS T1534 model in representing CCEWs and associated precipitation variability during the boreal summer, using NOAA OLR and NASA IMERG observations as reference across Day 1, 3, and 5 lead times. The model is able to reproduce the broad spatial distribution of tropical precipitation zones but tends to overestimate precipitation, with an increasing bias as the forecast lead time increases. An overestimation of light and moderate rainfall and an underestimation of heavy rainfall at extended lead times reflects limitations in the representation of convective processes and the partitioning between convective and stratiform rainfall, which are known challenges in global atmospheric models. At longer lead times, the model tends to underestimate the high frequency convective variability, indicating reduced skill in capturing synoptic scale precipitation bursts, which points to the need for improved representation of convection and sub-grid scale processes, as well as potential benefits from enhanced spatial resolution.

For CCEWs, the model reproduces several structural features of Kelvin waves but substantially underestimates their amplitude across the global tropics through all lead times. Weak easterlies over the Western Pacific likely contribute to reduced Kelvin wave activity, reflecting limitations in convection-large scale circulation coupling. ER wave variance is broadly captured but overestimated in magnitude—a bias that, consistent with earlier studies (e.g., DeMott et al., 2011), may affect the northward propagation of the boreal summer intraseasonal oscillation. The MJO exhibits weak eastward propagation throughout, indicating persistent deficiencies in representing the moist processes that sustain and propagate its convective envelope. Domain-averaged RMSE analysis confirms a

systematic increase in forecast error from Day-1 to Day-5, providing quantitative evidence of tropical variability degradation with lead time.

A key contribution of this study is the systematic assessment of the lead-time evolution of CCEWs variability in an operational high resolution forecasting system. The results highlight the sensitivity of convectively coupled disturbances to the representation of convection, moisture-convection coupling, and sub-grid scale processes, with forecast skill degrading progressively from Day-1 to Day-5. Improvements in these aspects, alongside increased spatial resolution, are likely necessary to better capture precipitation intensity and CCEW characteristics. Given the role of tropical waves in modulating monsoon variability and rainfall over the Indian region, accurate representation of CCEWs remains an important priority for operational forecasting and sub seasonal prediction.

Future work will extend this analysis using the recently developed high resolution version of the model (Bharath Forecast System model), with emphasis on the vertical structure and dynamical characteristics of CCEWs and their role in tropical precipitation variability.

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Author Contributions P. Mukhopadhyay, and H. Himabindu conceptualized the work. Material preparation, data collection and analysis were performed by H. Himabindu. The first draft of the manuscript was written by H. Himabindu. Review and editing done by P. Mukhopadhyay and all coauthors. The final

manuscript has been approved after integrating all feedback and suggestions from the coauthors.

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Data Availability

The datasets generated during and/or analysed during the current study are available in the “Pratyush” high performance computing system at IITM, Pune, India and can be made available by request to the corresponding author.

Declarations

Conflict of interest The authors declare no competing interests.

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